

Bouncing Back From Life's Perturbations: Formalizing Psychological Resilience From a Complex Systems Perspective

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Experiencing stressful or traumatic events can lead to a range of responses, from mild disruptions to severe and persistent mental health issues. Understanding the various trajectories of response to adversity is crucial for developing effective interventions and support systems. Researchers have identified four commonly observed response trajectories to adversity, from which the resilient is the most common one. Resilience refers to the maintenance of healthy psychological functioning despite facing adversity. However, it remains an open question how to understand and anticipate resilience, due to its dynamic and multifactorial nature. This article presents a novel formalized framework to conceptualize resilience from a complex systems perspective. We use the network theory of psychopathology, which states that mental disorders are self-sustaining endpoints of direct symptom–symptom interactions organized in a network system. The internal structure of the network determines the most likely trajectory of symptom development. We introduce the resilience quadrant, which organizes the state of symptom networks on two domains: (1) healthy versus dysfunctional and (2) stable versus unstable. The quadrant captures the four commonly observed response trajectories to adversity along those dimensions: resilient trajectories in the face of adversity, as well as persistent symptoms despite treatment interventions. Subsequently, an empirical illustration, by means of a proof-of-principle, shows how simulated observations from four different network architectures lead to the four commonly observed responses to adversity. As such, we present a novel outlook on resilience by combining existing statistical symptom network models with simulation techniques.

Keywords: resilience, complex systems, simulation modeling, network analysis, psychopathology

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Exposure to stressful or traumatic events can trigger a wide range of responses in individuals, from mild disruptions in functioning to severe and persistent mental health issues. Understanding the different trajectories of such varying responses to adversity is essential for developing effective interventions and support systems for those in need. Bonanno (2004) proposed that responses to adverse events typically follow four distinct response trajectories, including *resilience*, *recovery*, *delayed onset*, and *chronic dysfunction* (Figure 1). These trajectories were recently confirmed in a large classification study consisting of 54 studies (Galatzer-Levy et al., 2018). Resilience, the most prevalent trajectory type, is characterized by individuals who experience mild to moderate disruptions in normal functioning (e.g., a few weeks of disrupted sleep, occasional nightmares, or periodic uneasiness) but ultimately maintain a stable trajectory of healthy functioning (Bonanno, 2004). Other individuals experience a chronic dysfunction trajectory of enduring disruptions in functioning, a recovery trajectory of acute symptom elevations that gradually abate, or a delayed trajectory of mild to moderate symptoms that gradually worsen over time.

While it is important to understand the different trajectories that individuals exhibit in response to stress and adversity, a more comprehensive understanding of resilience requires investigating the underlying mechanisms of resilience that explain these individual differences. However, resilience is associated with a vast variety of variables across different domains, making it a difficult concept to define, let alone investigate. Studies have identified protective factors such as self-compassion (MacBeth & Gumley, 2012), social support (Gariépy et al., 2016), and positive emotions (Tugade & Fredrickson, 2004) that help individuals bounce back from adversity. But no set of key predictors that explain all differences in response trajectories after adverse events has ever been found (Bonanno, 2021; Modesto-Lowe et al., 2021).

Moreover, protective and risk factors do not operate in isolation but interact with each other (Ellis et al., 2017; Fritz et al., 2018; Riley

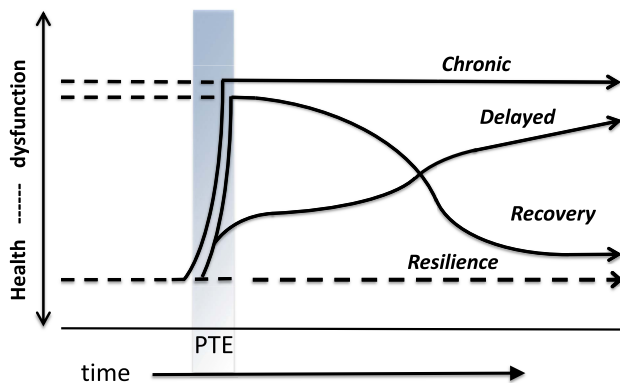
& Masten, 2005; Weinans et al., 2021; Xu & Kajikawa, 2018). For example, a study by Gijzel et al. (2019) found that, although both social support and physical activity were associated with better mental health in older adults, the protective effects of physical activity were stronger for those who also had higher levels of social support. In other words, the interaction between physical activity and social support was more important for mental health than the sum of the two factors alone. Furthermore, resilience is not a static trait but rather depends on a dynamical process that changes over time based on the stressors that an individual faces (Bonanno, 2004; Cramer et al., 2020; Hill et al., 2018; Kalisch et al., 2017; Rutter, 2012; Sapienza & Masten, 2011; Shafi et al., 2020; Stainton et al., 2019). Thus, resilience involves complex and dynamic interactions between risk and protective factors as well as the stressors that a person faces, which current resilience models fail to accommodate (Kalisch et al., 2019).

In recent years, several authors across disciplines have advocated for adopting a complexity perspective to conceptualize psychological resilience (e.g., Borsboom, 2017; Hasselman, 2023; Hill et al., 2018; Kalisch et al., 2019; Pincus & Metten, 2010). Within this complexity framework, resilience is not merely a variable within a system of multiple factors; instead, it emerges as a consequence of the system's inherent structure. A system can transition between various states, such as functional or dysfunctional state, and the *low propensity to change* from one state to another is termed resilience.¹ This conceptualization of resilience from complexity science integrates the literature's findings that no set of key factors can singularly explain resilience. Instead, resilience is one potential type of behavior of a complex system comprising a multitude of factors, and the system and its constitution can evolve over time or within different contexts, emphasizing the dynamic nature of resilience.

However, it remains an open question how to empirically study these conceptual ideas on resilience from a complexity perspective. A formal model grounded in empirical data is lacking, hindering progress not only in the conceptualization but also in the empirical examination of resilience from a complexity standpoint. In this article, we introduce the first formalized model that elucidates the dynamics of our focal phenomenon within the context of resilience: the four commonly observed response trajectories to adversity (Bonanno, 2004; Galatzer-Levy et al., 2018). Drawing inspiration from the network theory of psychopathology (Borsboom, 2017; Borsboom & Cramer, 2013), we posit that the four frequently observed response trajectories to adversity can be construed as distinct states within network models of interconnected mental disorder symptoms.

Through computational simulations, we subject diverse network architectures to adversity, illustrating how their behavior can be organized based on their projected state (healthy or dysfunctional) and stability (stable or unstable) against stressors or interventions. Consequently, this innovative approach paves the way for further

Figure 1
The Four Commonly Observed Response Trajectories to Adversity



Note. Four typical response trajectories to potentially traumatic events (PTE), including *resilience*, *recovery*, *delayed onset*, and *chronic dysfunction*. Adapted from “Loss, Trauma, and Human Resilience: Have We Underestimated the Human Capacity to Thrive After Extremely Aversive Events?” by G. A. Bonanno, 2004, *American Psychologist*, 59(1), p. 21 (<https://doi.org/10.1037/0003-066X.59.1.20>). Copyright 2004 by the American Psychological Association. See the online article for the color version of this figure.

¹ In this article, we align with the vast majority of the psychological literature and use the term “resilience” to describe a situation of positive mental health (e.g., see the reviews by Masten et al., 2021; Rutter, 2012; Ungar & Theron, 2020). Note, however, that in a complex science perspective, resilience is not necessarily restricted to describe the propensity to change of one particular state. Chronic depression reflects an ongoing struggle with poor mental health; nonetheless, in theory this pattern could also be viewed as viciously “resilient” to change in the sense that it is stable and not responsive to treatment interventions (see also Hill et al., 2024; Mahdiani & Ungar, 2021).

investigations and conceptualizations of resilience from a complex systems perspective in psychology, offering a practical avenue while integrating crucial findings from the resilience literature. The network theory of psychopathology proposes that mental disorders act as complex systems, which consist of interacting symptoms that influence one another (Borsboom, 2017; Borsboom & Cramer, 2013). In this view, mental disorders emerge from direct interactions among symptoms. For example, in the case of depression, someone experiencing insomnia may also experience fatigue, which can in turn lead to difficulties in concentrating. This can trigger a depressed mood and feelings of worthlessness, which ultimately results in a depressive episode. Such a depressive episode can become *self-sustaining* when the components of the system become locked into a stable state of dysfunction. This can occur through a process called reinforcement, in which the symptoms or behaviors associated with the disorder become mutually reinforcing and self-perpetuating (Scheffer et al., 1993). In other words, certain symptoms or behaviors may contribute to the persistence of other symptoms, which then in turn reinforce the initial symptoms or behaviors, leading to a self-sustaining cycle. For example, feelings of hopelessness and low self-worth may lead to social isolation, which then reinforces the feelings of hopelessness and low self-worth, perpetuating the depressive state. Once a self-sustaining state is established, it can be difficult to break out of the cycle of dysfunction. Thus, a mental disorder is viewed as a state where symptoms maintain each other, resulting in a persistent, dysfunctional pattern of behavior (McNally, 2012).

The network theory proposes that these mental disorders can be represented by *networks* in which symptoms are conceptualized as nodes and the interactions between them as edges. Network structures are estimated from empirical data using statistical network models (e.g., see Epskamp et al., 2018; C. D. van Borkulo, Borsboom, & Epskamp, 2014). As such, researchers can gain insights into how the network is structured. These network models contain node and edge parameters that represent how easily symptoms become present (node parameter: *threshold*) and how strong the associations between two specific symptoms are (edge parameters; Epskamp et al., 2018). The behavior of a symptom network is largely determined by these node and edge parameters, which together constitute the network's *architecture*. A simulation study found that strongly connected symptoms are more likely to activate each other (Cramer et al., 2016). For instance, if insomnia is strongly connected to excessive worrying, the presence of insomnia can increase the likelihood that excessive worry will also activate and vice versa. Additionally, each symptom has a threshold parameter for activation, with symptoms with low thresholds (like insomnia) being more easily triggered by stressors than those with high thresholds (like suicidal ideation). Thus, differences in network architecture may help explain individual differences in the development of mental health disorders (Schweden et al., 2018).

Symptom networks could also be extended with risk and protective factors that help or hinder the further development of symptoms. These risk and protective factors may directly affect the symptom network's architecture (Isvoranu et al., 2017; Kalisch et al., 2019; Lunansky, van Borkulo, et al., 2021). For example, the risk factor of being sexually abused in childhood is directly linked to the development of feelings of guilt later in life (Isvoranu et al., 2017), while the protective factor "social support" may hinder the evolution of "depressed mood" (e.g., Gariépy et al., 2016). Risk and protective

factors may directly influence the development of symptoms and alter their connections or thresholds. Protective factors, such as relationship quality or partnership satisfaction, may dampen the interactions between symptoms. When a person starts feeling worried, partners, family, or friends might decrease the probability that worrying will lead to other symptoms by offering social support (Røsland et al., 2012). Network architectures that are strongly influenced by the presence of many risk factors may thus exhibit stronger connections between symptoms and/or lower thresholds for developing symptoms, rendering them more vulnerable to stressful events. That is, in such vulnerable networks, stressors could lead to vicious cycles of persistent symptom activation resulting in full-blown depressive episodes in which the network gets stuck (Cramer et al., 2016).

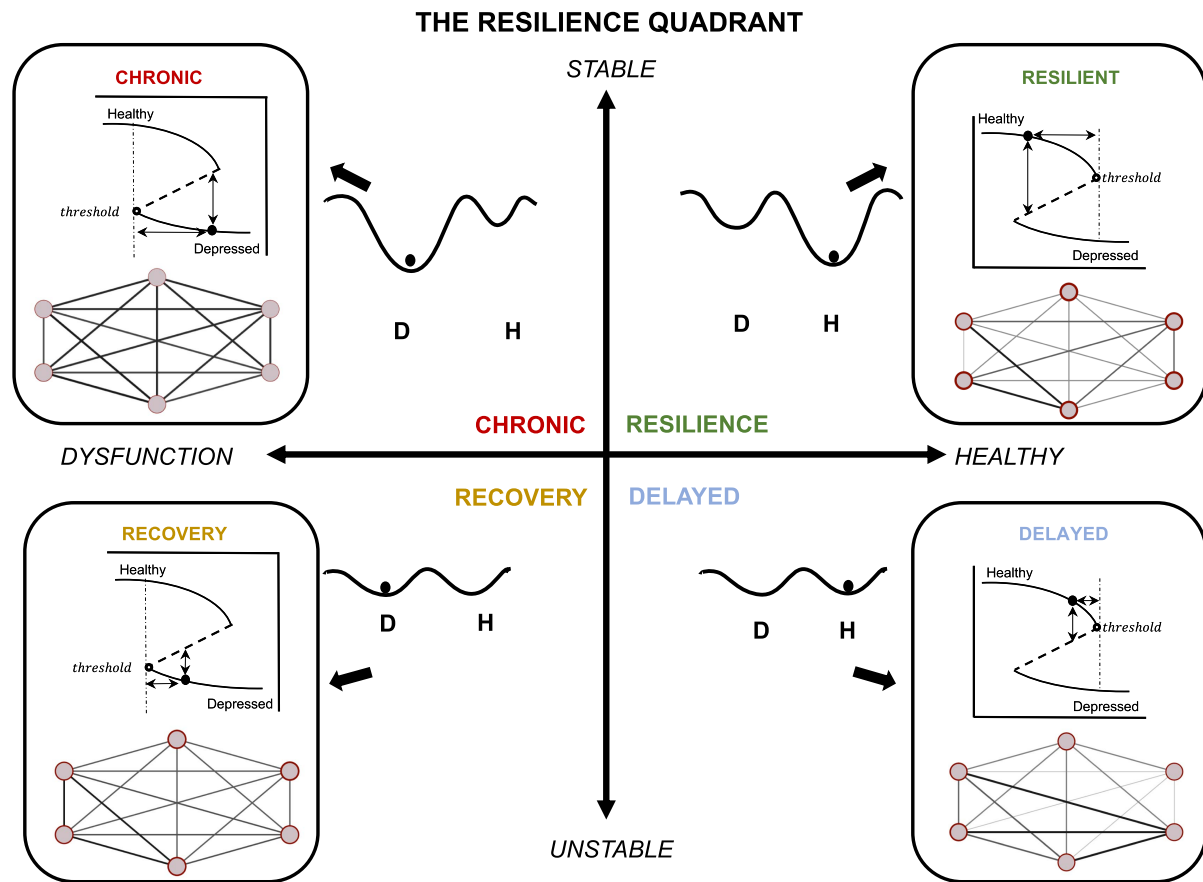
The article is set up as follows. First, we introduce the *resilience quadrant*, which organizes symptom networks based on their most likely *state* (healthy or dysfunctional) and *stability* (stable or unstable) against stressors or interventions. The quadrant captures different behaviors along these dimensions: resilient, recovery, and delayed trajectories in the face of adversity, as well as persistent symptoms despite treatment interventions (see Figure 2). Next, we give an empirical illustration showing that specific symptom networks (the Ising model; C. D. van Borkulo, Epskamp, et al., 2014) with exemplary architectures can be placed in different regimes of the quadrant, using a simulation framework. Last, we conclude with the necessary steps to include more insights from the resilience literature to further advance our approach.

The Resilience Quadrant

In this article, we focus on four types of trajectories (Figure 1) derived from a vast body of studies that mapped adjustment after exposure to highly aversive events that are typically associated with at least some impairment (e.g., bereavement). We present the resilience quadrant as a framework that organizes these four commonly observed response trajectories to adversity into four different regimes in which a symptom network can be located (see Figure 2). The quadrant organizes networks by two dimensions: the network's *projected state* (state-axis) and the *stability of that state* (stability-axis). The state-axis denotes whether the system can be expected to be in a healthy or dysfunctional state, given the network's architecture, in other words, how many symptoms are expected given the network's architecture which is *simulated from the current reported level of symptoms under normal circumstances*. The stability-axis indicates how susceptible the projected state is to change when faced with stressors or interventions, called *perturbations*. Using the four commonly observed response trajectories to adverse events, a network of depression symptoms can be located in a resilient, delayed onset, recovery, or chronically dysfunctional regime of the quadrant. We define resilience specifically as an outcome in which no transition into a dysfunctional state takes place, despite facing these highly aversive events.

A visual metaphor that can explain the discussed link between the stability of the network's state and resilience is the ball-in-a-cup metaphor (see Figure 3, which is often used in dynamical systems theory; e.g., see Scheffer et al., 2018). The cup represents the *attractor basin* of the network, which corresponds to the state the network is gravitating toward. As introduced above, the states are simplified to two options: a healthy state with few active (i.e., present) symptoms and a dysfunctional state with many active symptoms. The ball represents the system's current state, which can take on all

Figure 2
The Resilience Quadrant



Note. The center shows the resilience quadrant of a (simplified) mental health system with two alternative stable states: healthy and dysfunction. The valleys in each quadrant represent the underlying attractor landscape for the two alternative states, in which “D” stands for the depressed state and “H” for the healthy state. For example, in the “resilient” regime, the system is located within the deep, steep, and wide attractor basin of the healthy state, and therefore it is unlikely to fall into the shallow attractor basin of the dysfunction state, even in the face of adversity. The recovery and delayed stability landscapes show unstable attractors for both states, depicting that these trajectories show more fluctuation from one state to another. The corresponding boxes show the current location in the system’s trajectory, where the upper trajectory represents the healthy state and the lower trajectory represents the dysfunction state. Within each box, corresponding network architectures are shown. Nodes represent depression symptoms, and edges represent associations (e.g., logistic regression coefficients) between symptoms, while the thickness of edges represents the strength of this relationship. The thickness of red borders around nodes represents the magnitude of threshold parameters, which differ per node. See the online article for the color version of this figure.

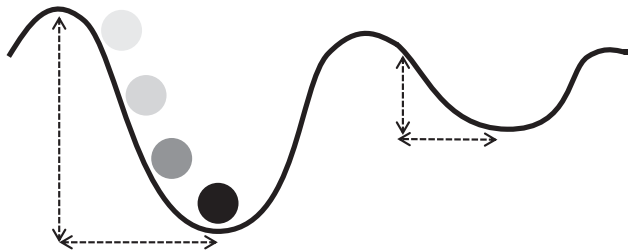
possible combinations of active and inactive symptoms. The steepness and width of the cup represent how stable the ball lies in the attractor. If the attractor basin is broad, deep, and steep, the ball is likely to stay at the lowest point and unlikely to move and be pushed over the edge. In contrast, an attractor basin that is narrow, shallow, and flat is unstable since the ball is likely to roll out of the attractor basin.

The ball can be pushed by perturbations, such as minor daily stressors or adverse life events (e.g., divorce or the death of a loved one; [Fried et al., 2015](#)). However, the external events may also represent alleviating interventions, such as clinical treatments that aim to move individuals toward a healthy state. The influence of the perturbation on the ball’s behavior depends on the force pushing the ball forward (i.e., how strong the perturbation itself is) and the form

of the cup (i.e., how deep and steep the attractor is). Together, these determine the probability that the ball will switch from one state to another. For example, a seriously adverse situation, such as a natural disaster or car accident, will push the ball forward with unusual force, while a minor daily stressor could only slightly tip the ball forward.

It is important to note that the current landscape that determines the state in which networks will be located is inherently flexible ([Olthof et al., 2023](#)). The landscape could change across situations or over time as individuals have novel experiences. Changes in one’s environment, mindset, and personal situation would lead to different interactions between risk factors, protective factors, and symptom interactions, meaning that the network model and underlying stability

Figure 3
Ball-In-A-Cup Metaphor



Note. The depth and width of the attractor determine how easily an imaginary ball falls into and out of the attractor after being pushed and, thus, how stable the system is currently located in the attractor. The left basin is stable, represented by its wide and steep valley. In contrast, the right basin is unstable due to its shallow and narrow form.

landscape would also change. In other words, individual parameters of the symptom network model and the underlying stability landscape could change, meaning that resilience levels may evolve over the lifespan and change across situations. The position of the network model in the landscape is, therefore, only reflective of the current situation.

A network situated in a healthy and stable attractor, with an unstable attractor for the dysfunctional state, will show a resilient trajectory. This means that resilience is no longer a specific component of the network (e.g., a variable in the network or a particular set of factors) but a *property* of the network.

In the resilient regime (the upper right area of Figure 2), the network is currently located in a healthy and stable state. Exposure to an aversive event may lead to the development of some symptoms (e.g., feelings of loneliness after the death of a spouse), but this small number of symptoms will not further evolve toward a phase transition into a dysfunctional state. Additionally, the resilient regime is also characterized by positive aspects of adjustment. In this article, we only focus on symptom trajectories, but other studies that include positive metrics such as life satisfaction and well-being have established a link between little or no symptoms and indications of positive adjustment in individuals characterized by a resilient trajectory (Bonanno et al., 2024). Individuals within this healthy and stable regime may *flexibly* utilize strategies and resources that allow them to cope effectively with high levels of distress and uncertainty (Bonanno & Burton, 2013; Bonanno et al., 2024). For example, bereaved individuals showing few symptoms of depression also report feelings of comfort from having positive memories about their deceased loved one (Bonanno et al., 2004). Networks with weak connectivity and high thresholds will be located in this regime. The healthy state can also be unstable in which more symptoms are easily activated after facing perturbations, which is depicted in the lower right regime (“delayed onset”) of the quadrant (see Figure 2). Individuals currently characterized by this network typically show healthy functioning, but they increasingly develop more mental health problems after facing stressful events until they can reach a dysfunctional state. Networks with slightly stronger connectivity but lower thresholds than the resilient network are located in the delayed regime. The left part of the quadrant depicts the dysfunctional state of the network, in which the upper regime depicts the chronic dysfunction

regime. A network in this regime is stuck in a dysfunctional state after a stressful event and unresponsive to clinical interventions. Individuals who suffer from chronic depression can be hypothesized to fit this category, as they are often not responsive to many (first) treatment interventions (Fava, 2003). Networks with strong connectivity and low thresholds will be located in this regime. The dysfunctional state can also be unstable, which is represented by the “recovery” trajectory in the bottom-left regime of the quadrant. Individuals in this regime develop a large number of mental health problems after facing adversity, but they respond well to treatments or may recover on their own without treatment. The recovery networks also have strong connectivity and low thresholds, but these parameters are smaller than in the chronic dysfunction network.

Proof-of-Principle

In the current section, we present an empirical illustration of the proposed resilience framework as a proof-of-principle. The theoretical framework could, in principle, be applied to many different types of (network) models, and with this empirical illustration, we aim to demonstrate the framework’s practical potential by showing a concrete example with a specific network model. The proof-of-principle illustrates how an adapted version of the Ising network model (C. D. van Borkulo, Epskamp, et al., 2014) can show dynamics that correspond with the four commonly observed trajectories to adversity (Galatzer-Levy et al., 2018) and align with the four regimes in the resilience quadrant.

The setup of the proof-of-principle is as follows. To make sure that the constructed networks are somewhat bound to empirically estimated parameter values, we first estimate a symptom network from empirical data containing assessments of nine major depressive disorder (MDD) symptoms as our baseline model. Next, we show how alterations in the architecture of this baseline model correspond with behavior from the four commonly observed response trajectories to adversity (Galatzer-Levy et al., 2018). We construct four different symptom network models by altering the architecture of estimated baseline model to mimic characteristic behavior for each regime of the resilience quadrant (Figure 2) corresponding with the four commonly observed response trajectories. The aim of constructing these networks was to show, as a proof-of-principle, how four typical network architectures correspond with the four response trajectories. For each of these four constructed networks, we simulated novel observations that show if the network model propels toward a healthy state (low number of symptoms) or dysfunctional state (high number of symptoms). Afterward, we added perturbations to these simulated observations to assess how stable the projected state is: The resilient network model should show no significant increase in projected symptom activity despite facing adversity, while the network model representing the chronic dysfunction trajectory would be insensitive to treatment interventions.

All analysis code and research materials are available at <https://osf.io/gv46e/>. Analyses were executed with R, Version 4.1.2 (R Core Team, 2021) and the packages *IsingFit*, Version 0.3.1 (C. D. van Borkulo, Epskamp, et al., 2014); *qgraph*, Version 1.9 (Epskamp et al., 2012); *ggplot2*, Version 3.3.5 (Wickham, 2016); *tidyverse*, Version 1.3.1 (Wickham et al., 2019); and *ggpubr*, Version 0.4.0 (Kassambara, 2020). This study’s design and its analysis were not preregistered.

The Baseline MDD Model

Data

The baseline symptom network is estimated from data from the Virginia Adult Twin Study of Psychiatric and Substance Use Disorders (Kendler & Prescott, 1999) and contain binary data on the presence/absence of nine MDD symptoms from 8,973 twins from the Midatlantic Twin Registry. The symptoms are: (a) depressed most of the day, nearly every day; (b) markedly diminished interest or pleasure in all, or almost all, activities most of the day, nearly every day; (c) significant weight loss when not dieting or weight gain (e.g., change of more than 5% of body weight in a month), or decrease or increase in appetite nearly every day; (d) insomnia or hypersomnia nearly every day; (e) psychomotor agitation or retardation nearly every day; (f) fatigue or loss of energy nearly every day; (g) feelings of worthlessness or excessive or inappropriate guilt nearly every day; (h) diminished ability to think or concentrate, or indecisiveness, nearly every day; and (i) recurrent thoughts of death, recurrent suicidal ideation without a specific plan, or a suicide attempt or a specific plan for committing suicide. Symptoms were measured during a psychiatric interview using an adaptation of the Structured Clinical Interview for *Diagnostic and Statistical Manual of Mental Disorders*, 3rd ed., revised (Spitzer et al., 1992).²

Model

We estimated the baseline symptom network model using an adaptation of the Ising model (C. D. van Borkulo, Epskamp, et al., 2014). The Ising model is a relatively simple model that has been frequently applied to mental health, specifically depression, for its ability to show complex behavior in line with the development of psychiatric disorders. For example, the Ising model can show *nonlinear transitions* (e.g., sudden changes in mental health complaints) and *hysteresis* (the persistence of mental health complaints even when initial stressors have abated; see Cramer et al., 2016; Finnemann et al., 2021; Marsman et al., 2018; C. D. van Borkulo, Borsboom, & Epskamp, 2014, for an extensive description of the Ising model used in psychology). Symptoms are encoded as either “present” (1) or “absent” (0). The Ising model is symmetric, meaning that the relationships between the nodes are undirected; for example, the relationship from “insomnia” to depressed mood is the same as the relationship from depressed mood to insomnia.

Constructing the Four Network Models

We adapted the baseline network model in four ways that show the dynamics of the four commonly observed responses to adversity. To do so, we multiplied the connectivity (edge) and threshold (node) parameters with certain constants in order to maintain the same relative structure of the baseline network in every adaptation (see Figure 4). We searched for the smallest changes in baseline parameter magnitudes that showed the dynamics of each of the response trajectories. The positive or negative direction in which the connectivity and threshold parameters should be changed were garnered from earlier simulation studies on the Ising model (Cramer et al., 2016) and the mathematical formula that governs the Ising model dynamics (see the Supplemental Materials).

To construct the network in the resilient regime, we weakened the connectivity by multiplying the baseline edge weight parameters

with 0.8 and made the thresholds higher (i.e., the symptoms will be less likely to activate) by multiplying the baseline threshold parameters with 1.2, which will make the healthy state of the network more stable. We did the opposite for the network located in the chronic dysfunction regime, increasing the connectivity by multiplying the baseline edge weight parameters with 1.2 and decreasing the thresholds (i.e., making the symptoms more likely to activate) by multiplying the baseline threshold parameters with 0.8. We made the network dynamics more unstable for the delayed and recovery networks by altering the connectivity and threshold parameters in opposite directions. For the delayed response network, we multiplied the edge weights of the baseline model with 1.1 and the threshold parameters of the baseline model with 0.95. For the recovery network, we multiplied the edge weights of the baseline model with 1.1 and the thresholds with 0.9.

Simulating Observations From the Network Models: Checking the Projected State

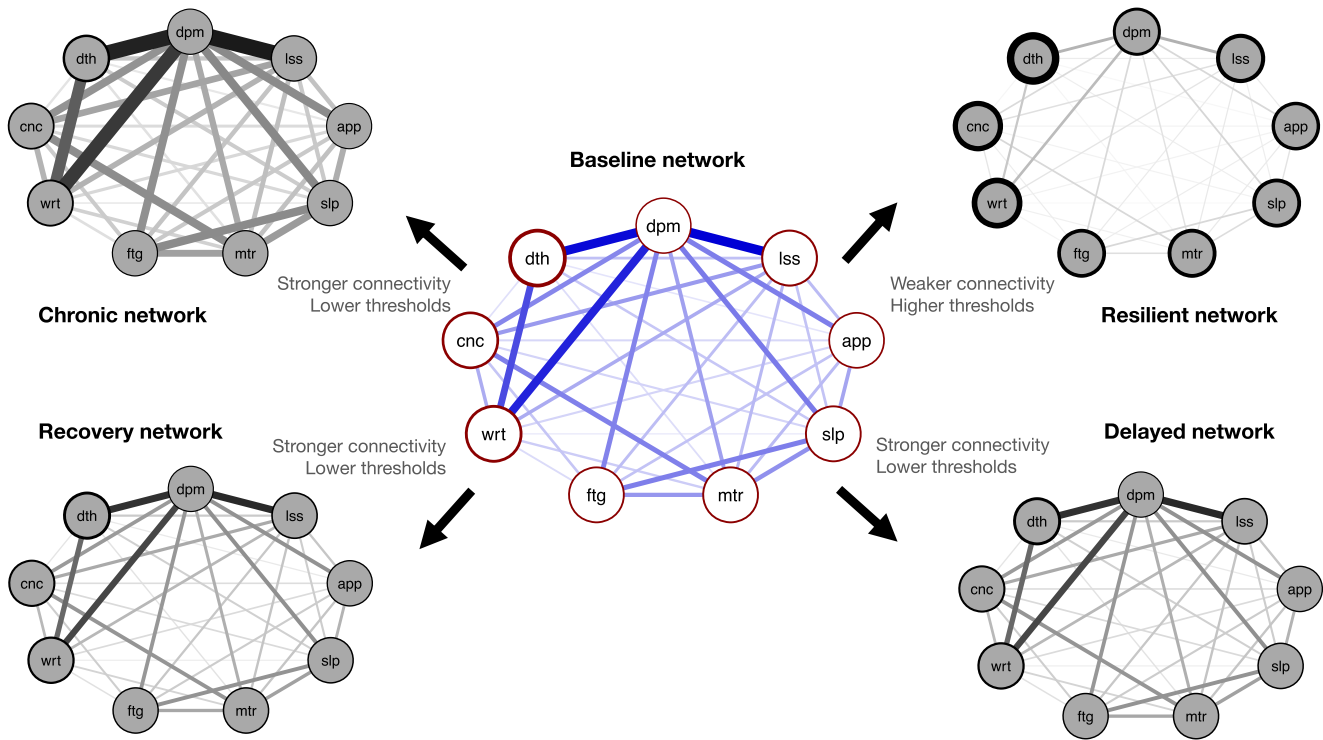
First, we determine the state (healthy or dysfunctional) that the networks are pulled toward given the network architecture, meaning that there are no perturbations. We simulated the expected symptom dynamics of the four different network models to show how these four different patterns match the four regimes in the resilience quadrant. To study the projected state of the network models, we simulated many observations from the network using Glauber dynamics³ (Glauber, 1963). The Supplemental Materials explain the simulation technique used to simulate the observations from the networks in detail. The simulated observations can be seen as “repeated measures” of the nine depression symptoms for the same fictitious population of participants that each network represents. In that way, the simulated observations approximate the most likely symptom developments of every network, given the network’s architecture. We calculated the projected state of every network by taking the mean of the sum scores of active symptoms over all simulated observations. The mean sum score falls below or above a specific clinical cutoff value for diagnosis. The clinical cutoff value used in these simulations for the current proof-of-principle is the (simplified⁴) clinical cutoff value for diagnosing MDD, namely, a sum score of five or more active symptoms denoting a dysfunctional state and less than five active

² The Structured Clinical Interview for *Diagnostic and Statistical Manual of Mental Disorders*, 3rd ed., revised, which was the most recent instrument at the time of the original data collection from which our baseline model is estimated (Spitzer et al., 1992). In this article, we are mostly interested in showing how changes in the network architecture, specifically in the connectivity strength and threshold parameters, lead to different response trajectories to adversity. The main goal of estimating the baseline network is to ensure that the estimated relations between symptoms are rooted in empirical data (contrary to assuming a random structure of a symptom network).

³ For specific types of network models, the most likely state of a network can also be calculated analytically. For example, the most likely state of the model used in this article (the Ising model) can be solved analytically for networks up to 10 nodes using the R-package *IsingSampler* (Epskamp, 2020). However, as we also simulate perturbations to check the stability of the most likely state, we use the same type of simulations to position the network on both the state- and stability-axis.

⁴ The *Diagnostic and Statistical Manual of Mental Disorders*, 5th ed. (American Psychiatric Association, 2013) states that for a diagnosis of MDD, five symptoms should be present, of which “loss of interest” or depressed mood should be present (American Psychiatric Association, 2013). For simplicity, here, we do not consider which specific symptoms are present.

Figure 4
Constructing Network Models Representing Four Commonly Observed Response Trajectories to Adversity



Note. We created four networks that show resilient, delayed, recovery, and chronically dysfunctional trajectories by manipulating an empirical baseline network model (in the center). The four networks were created by multiplying the connectivity and threshold parameters of the baseline Ising network model's architecture and, in this way, increasing or decreasing the parameters (the changes are dramatized in the figure for visual clarification). The four constructed networks show behavior that fits the four regimes of the resilience quadrant. The thickness of edges represents the magnitude of the connectivity, and the thickness of node borders represents the strength of the thresholds. As can be seen from the figure, the delayed and recovery networks have quite similar architectures. Abbreviations of the nodes refer to depression symptoms as follows: dpm = depressed mood; lss = loss of interest; app = appetite disturbance; slp = sleep disturbance; mtr = psychomotor agitation; ftg = fatigue; wrt = feelings of worthlessness; cnc = concentration problems; dth = thoughts of death. See the online article for the color version of this figure.

symptoms indicating a healthy state. Importantly, in our proposed framework, the cutoff value between the healthy and dysfunctional state is determined by the researcher based on the research question, the symptoms that are being studied, and the context of the study. Table 1 show the results from the simulations. The projected state for the *resilient network* is healthy since the mean sum score of simulated observations falls well below the cutoff of five symptoms ($M = 0.04$, $SD = 0.19$). The *delayed network* showed more

fluctuations. There is no clear state for this model, and the number of active symptoms switched from a few active symptoms to a state with a high number of active symptoms. However, based on the mean of the active symptoms ($M = 3.83$, $SD = 3.95$), the projected state of the networks falls below the cutoff value of five or more active symptoms. It is therefore located within the healthy regime of the resilience quadrant. The *recovery network* showed some fluctuating symptom evolutions but with more active symptoms. The mean sum

Table 1
Simulated Dynamics With and Without Perturbations for All Constructed Networks

Symptom sum score	Constructed symptom network							
	Resilient		Delayed		Recovery		Chronic dysfunction	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Without perturbations	0.04	0.19	3.83	3.95	8.36	0.87	8.87	0.47
With perturbations	0.19	0.85	5.76	3.65	4.13	4.1	7.41	3.22

score of the recovery network falls above the cutoff value ($M = 8.36$, $SD = 0.87$), and its projected state is therefore dysfunctional. The *chronic dysfunction network* model showed many active symptoms ($M = 8.87$, $SD = 0.47$), and its projected state is consequently dysfunctional.

Simulating Perturbations: Checking the Stability

After evaluating whether the characteristic behavior exhibits healthy or dysfunctional states, we assessed the second axis of the resilience quadrant: stability versus instability. To this end, we perturbed the networks to study how stable their projected state is. Note that different perturbations are needed to examine the stability of a healthy or dysfunctional projected state. To study the stability of a healthy projected state, we apply *aggravating interventions* mirroring the development of depressive symptoms due to an adverse life event. Aggravating interventions temporarily increase the level of symptoms in the simulations. To the contrary, we apply *alleviating interventions* to investigate whether a disordered projected state is susceptible to change. Alleviating interventions represent treatment interventions by forcing symptoms to be inactive at specific moments of the simulation. Figure 5 gives an example of simulated aggravating interventions, and the Supplemental Materials explain the simulations in detail. After the perturbations, the simulation of observations continued in its regular fashion. The network's architecture determines whether the simulated observations can get "stuck" in the absence or presence of symptoms. We calculated the effect of the interventions by computing the mean of the active symptoms over all observations. This symptom mean can again fall above or below the chosen clinical cutoff value, determining whether the system's projected state after facing aggravating or alleviating interventions remains healthy or dysfunctional.

Based on the previous results, we simulated aggravating interventions for the networks projected to be in the healthy state and alleviating interventions for the networks projected to be in the unhealthy state (See Table 1 and Figure 6). The resilient network bounced back very quickly from every aggravating intervention and remained in a healthy state with almost no actively present symptoms ($M = 0.19$, $SD = 0.85$). Interventions had more effect on the delayed network, in which the state of the network sometimes reached a dysfunctional state. However, symptom evolution patterns were still highly fluctuating. The mean sum score of the network now falls above the cutoff ($M = 5.76$, $SD = 3.65$). In the recovery network, the applied alleviating interventions successfully diminished the number of symptoms but with high fluctuations ($M = 4.13$, $SD = 4.1$). The chronic dysfunction network model remained dysfunctional most of the time despite alleviating interventions. However, the alleviating interventions seem to have some effect as symptom presence fluctuated ($M = 7.41$, $SD = 3.22$).

Determining the Position of the Networks in the Resilience Quadrant

We evaluated where the networks are located in the resilience quadrant by comparing the previous results. First, we assessed the networks on the health versus disorder domain to locate them on the left or right side of the state-axis in the quadrant (See Figure 2). If the network's projected regime fell below (above) the clinical cutoff, we positioned the network on the healthy (dysfunctional) domain in the right (left) side of the quadrant. Second, we evaluated the

networks on the stable versus unstable domain to locate them on the upper or lower side of the stability-axis in the quadrant. If the network's projected state after perturbations remained the same as the state without perturbations (i.e., the mean sum score of active symptoms still falls below or above the clinical cutoff; healthy or dysfunctional), we positioned the network on the stable regime on the upper side of the quadrant. Otherwise, we located it in the unstable regime on the lower side of the quadrant.

Figure 6 shows the results of all the simulations for the four networks. The resilient network showed a healthy state, which was stable against aggravating interventions and is therefore located in the resilient regime of the resilience quadrant. Notably, some symptoms may occur after the adverse event, but on average, the expected dynamics after the perturbations stay well below the cutoff value of the dysfunctional state. The delayed network showed unstable dynamics, with a symptom mean score below the cutoff, but which jumped toward an unhealthy state after aggravating interventions. Therefore, the delayed network is located in the unstable healthy (delayed onset) regime. The recovery network is projected to be in an unstable dysfunctional state, which can transition to a healthy state after alleviating interventions and is therefore placed in the recovery regime of the quadrant. The chronic dysfunction network remained dysfunctional despite alleviating interventions and is thus situated in the resilience quadrant's stable dysfunctional ("chronic dysfunction") regime.

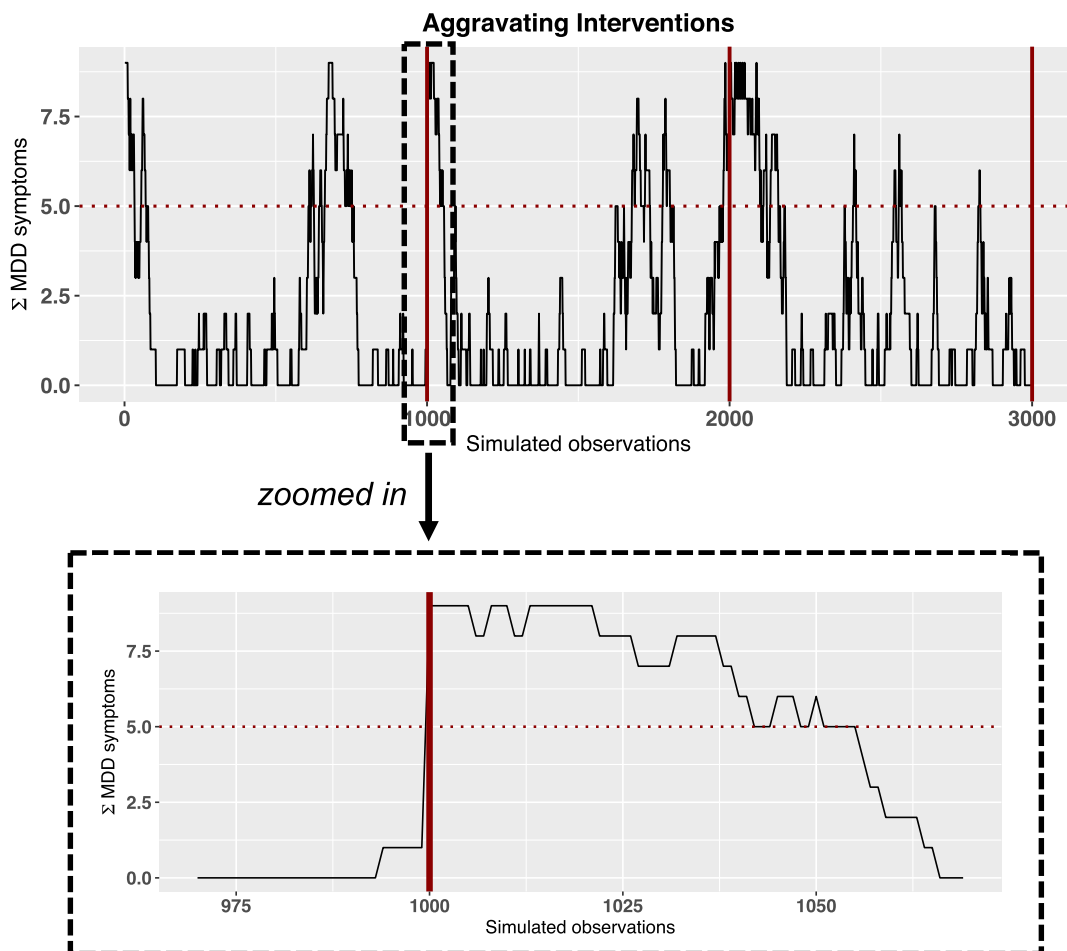
The Relation Between Network Architecture and Resilience

The networks in the four quadrants were constructed from a baseline, estimated network. We arrived at the typical networks by systematically altering the parameters of the baseline network. This process is represented in Figure 4. As is shown in the figure, a chronic dysfunction network arises when interactions between symptoms are strong (increased edge weights), and the tendency of each symptom to be active is stronger (lower thresholds). Thus, both stronger interactions with other symptoms and the degree to which symptoms have an autonomous tendency to be active contribute to a loss of resilience. In contrast, the resilient network is a perfect mirror of the chronic dysfunction network: it features weaker symptom-symptom interactions and a more favorable threshold of the individual nodes. Notably, the *delayed* and recovery networks are located in the parameter space in between the chronic dysfunction and resilient networks. They have slightly less favorable thresholds and somewhat higher edge weights. In this area of the parameter space, simulations suggest that many (nearly) equivalent models can be constructed through slight changes in the model parameters. This indicates that to assess resilience, the threshold parameters of the Ising model cannot be ignored, as lower thresholds can offset stronger interactions and vice versa. In addition, observe that small changes in the model's parameters suffice to alter their position in the resilience quadrant. We return to the implications of these results in the discussion.

Discussion

In this article, we proposed a novel formalized framework to conceptualize resilience using a complex systems approach. The framework organizes commonly observed responses to adversity

Figure 5
An Example of Simulated Aggravating Interventions



Note. Simulations are based on the estimated baseline symptom network model (see [Supplemental Materials Appendix B](#)). The x-axis represents the simulated observations, while the y-axis represents the sum scores of the active depression symptoms. The red vertical lines represent every moment an aggravating intervention is simulated, meaning that all symptoms are forced to be active for one moment in the simulation. The dashed red line denotes the (simplified) clinical cutoff value for diagnosing MDD, namely a sum score of five active symptoms. The lower figure is zoomed in on one intervention to show how the system recovers in detail. In this plot, the network with these estimated parameters generating these dynamics will fall below the clinical cutoff value for diagnosing depression most of the time. MDD = major depressive disorder. See the online article for the color version of this figure.

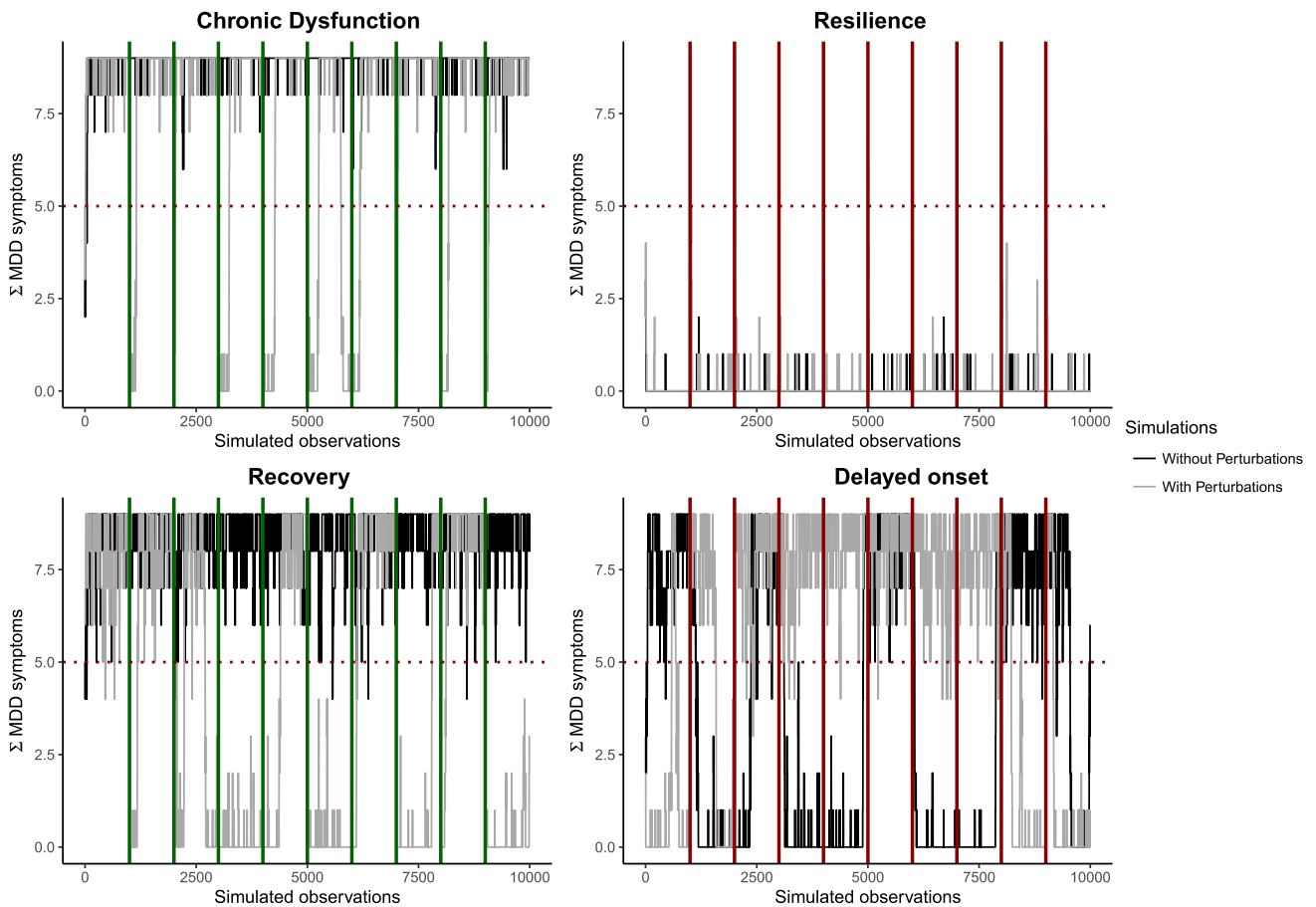
along two dimensions: the state the system is typically pulled toward (healthy or dysfunctional) and the stability of that state against perturbations. Individuals showing a resilient response to adversity are characterized by a mental health system that is located in a healthy and stable attractor, allowing them to maintain healthy functioning despite facing adversity. The framework builds upon the network theory of psychopathology (Borsboom, 2017; Borsboom & Cramer, 2013) and suggests that the mental health system could be represented by symptom network models. Different architectures of such a symptom network model would render different response trajectories to adversity.

As a proof-of-principle, we implemented the framework in a specific type of network model, the adapted Ising network model

(C. D. van Borkulo, Epskamp, et al., 2014), to showcase an empirical illustration of the practical potential of the framework. Specifically, we combined (estimated) symptom network Ising models with simulations to study how resilient a symptom network behaves in relation to its architecture. First, we assessed whether the system characteristically exhibits symptom activation beyond the clinical cutoff. Second, we evaluated whether the system state is robust against perturbations using simulations (where these are either aggravating or alleviating depending on the characteristic state of the system). Applying this analysis to a baseline symptom network of depression, an expected finding is that stronger connectivity and less favorable thresholds together can make a network chronically dysfunctional.

Figure 6

The Projected Symptom Dynamics for Four Networks in Distinct Regimes of the Resilience Quadrant



Note. The x-axis of every plot shows the simulated observations, and the y-axis shows the sum of the present symptoms, where a higher score indicates a higher level of dysfunction. The dashed horizontal red line indicates the clinical cutoff value of five symptoms or more for a diagnosis of MDD. The vertical lines indicate the moments where interventions are simulated, where red lines indicate aggravating perturbations (all symptoms are forced to be present), and green lines indicate alleviating interventions (all symptoms are forced to be absent). The upper-right plot shows the symptom evolutions for the symptom network located in the resilient regime. Despite aggravating perturbations, it displays a healthy and stable state, which follows from the projected low number of present symptoms. The lower-right panel shows the results for the network that was constructed to show an unstable healthy state located in the delayed onset regime of the quadrant. The projected state of the network is healthy, but it is heavily affected by the aggravating perturbations. The lower-left panel shows the projected evolutions of the network located in the recovery regime, which is characterized by an unstable dysfunctional state that recovers from alleviating interventions. The upper-left panel shows the results for the network located in the chronic dysfunction regime, which is characterized by high levels of symptomatology despite alleviating interventions. The chronic dysfunction network is therefore unlikely to be susceptible to treatment interventions. MDD = major depressive disorder. See the online article for the color version of this figure.

Two important additional implications emerged. First, the effects of changes in thresholds and connections can offset each other in the Ising model. This finding is important to incorporate in research. It implies that empirical tests, such as those based on the Network Comparison Test (C. D. van Borkulo et al., 2023), which compares networks based on only their connectivity, should not be confused with tests of resilience. This is because two networks with equal connectivity but different thresholds may show different levels of resilience. Second, the differences in parameter values that result in different quadrants are slight. This could clarify why resilience is believed to rely on complex configurations of supportive factors, where numerous factors interact with relatively weak connections to each other.

It is our hope that thinking about and studying resilience from a complex systems approach will ultimately lead to novel methods that can improve the assessment and prediction of response trajectories to adverse events or treatment efficacy. For example, this approach could help decide what the network structure of a patient may look like and, accordingly, help identify in which regime of the resilience quadrant patients may be located. This information could, in turn, contribute to determining optimal intervention strategies, as patients with more resilient network structures may need less strong interventions compared with patients with less stable symptom networks. In the remaining section, we outline the most critical steps that, in our view, will help advance the research program on resilience from a complex systems approach in psychology.

Extend the Ideas Presented in This Article to Other Statistical Network Models

The network model that was used in our simulations (see [Supplemental Materials Appendix B](#) for details) is helpful for theoretical exercises since it is a relatively simple model that can accommodate various complex phenomena aligning with mental disorders, such as phase transitions from healthy to dysfunctional states ([Finnemann et al., 2021](#)). However, the model used in this article originated in physics and was designed to explain ferromagnetism (see [Supplemental Materials Appendix A](#)). The model is potentially too simplified to be a realistic representation of the dynamic course of symptom development. For example, symptoms can only be active or inactive in the model, and the associations between symptoms cannot change over time. However, the ideas outlined in this article could be extended to other models that are more useful to represent the course of depressive complaints or resilient responses. The Gaussian graphical model ([Epskamp et al., 2018](#)) allows for continuous and ordinal data, such as symptoms measured on a Likert scale. The mixed graphical model ([Haslbeck & Waldorp, 2020](#)) is a network model that can be estimated from both continuous and categorical data and is thus a suitable candidate to study the relationships between symptoms and various types of risk or protective factors. Vector autoregressive models ([Bringmann et al., 2013](#)) can be used to study the relationships of variables over time, such as daily fluctuations of moods and affect states (e.g., see: [Snippe et al., 2017](#); [Wichers et al., 2015](#)). One could also add interventions to continuous-time vector autoregressive models ([Ryan & Hamaker, 2022](#)). Ideally, models would also be built from the ground up to allow for specific time-varying elements of resilience, such as the computational model for Panic Disorder ([Robinaugh et al., 2024](#)).

Expand the Modeling of Perturbations

In this article, we proposed a way to represent and study the resilience of symptom networks as an outcome by comparing simulated dynamics before and after alleviating or aggravating interventions. Perturbations could be extended such that they represent different types of adversity or treatment interventions in life. Instead of representing interventions as (de)activating all symptoms at specific moments, one could also model interventions that represent chronic stressors, minor daily hassles, or permanent treatments (i.e., taking medication that permanently impacts symptom development). One could, for example, (de)activate specific symptoms for the duration of the simulations. In addition, one could target specific symptoms for alleviating or aggravating interventions to investigate which symptom is projected to have the largest influence on the resilience level of the network ([Lunansky, Naberman, et al., 2021](#)). Alternatively, one could target a random number of symptoms for every intervention, realistically representing stressful life events or treatment interventions.

Include Risk and Protective Factors to Symptom Networks

In the current framework, risk and protective factors are theorized to alter symptom network architecture, and, in that way, resilience. One could study whether groups of people that score high on certain

risk or protective factors do indeed have a different symptom network structure or whether other factors, for example, flexible regulation, are also required ([Bonanno et al., 2024](#)). This could be achieved by estimating a network from data on both symptoms and risk and protective factors (using, e.g., the mixed graphical model network for mixed data; [Haslbeck & Waldorp, 2020](#)). Afterward, one may condition on specific values of the risk and protective factors, such as the presence of protective factors and the absence of risk factors, to obtain the symptom network structure for the people represented by that situation. One could feed that symptom network into the framework proposed in this article by simulating data and perturbations from the specific symptom network architecture. In this way, one could study whether specific risk and protective factors lead to differences in resilience from a complexity approach. One limitation, for now, is that the simulation techniques presented in this article only work for binary data. Future research could apply the proposed logic to networks estimated from different data, such as mixed data (i.e., symptoms and risk or protective factors measured on different scales).

Work on a Personalized Framework of Resilience Assessment and Prediction

Our presented framework may be useful in generating hypotheses on network models of different participant groups, such as comparing clinically depressed patients with healthy individuals. One could compare the networks of groups of people that did and did not develop mental disorders over time ([C. van Borkulo et al., 2015](#)). Using our proposed framework, an exciting possibility is to investigate whether differences between these groups might already be detectable at baseline. However, this would not mean that we could predict response trajectories or assess the resilience of every individual from these groups. One of the biggest challenges for formalizing psychological phenomena such as mental health is the heterogeneity of groups and diagnoses ([Molenaar, 2004](#)). Heterogeneity implies that trajectories at the group level do not directly translate to every individual ([Hamaker, 2012](#)). To personalize a formal framework of resilience, our proposed framework needs to be adapted to individual models. A recent example of a study in this direction suggests monitoring individuals over time on stressor exposure and mental health problems ([Kalisch et al., 2021](#)). Deviations of the individuals from a “normative” trajectory (e.g., based on the mean of the studied sample of participants) could be detected early and related to later outcomes in order to detect whether risk for developing problems exists. Another example is to combine intensive data collection of daily momentary moods and stressful events with periodic assessments of symptom evolutions ([Kuranova et al., 2020](#)). In this way, one could study if the recovery from adverse events, in terms of maintaining positive momentary moods, could be related to an increase in symptom development or resilient responses.

Conclusion

Resilience is an inherently multifactorial and dynamic concept. As such, we have to understand how different mechanisms work together to make a person cope, thrive, or break down after facing hardship ([Ioannidis et al., 2020](#); [Kalisch et al., 2017](#)). This article presented a novel way to conceptualize resilience from a complex systems perspective in psychology by combining existing statistical

symptom network models with simulation techniques. We hope that the general framework will aid future researchers from different domains to advance the study of resilience collaboratively.

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